

**Advanced Topics in Econometrics**  
**(PhD in Mathematics Applied to Economics and Management)**  
**2025/2026**

**Exercise Sheet 11 - GARCH Models**  
(version 23/11/2025)

1. Consider the ARCH(1) model,

$$\begin{aligned} Y_t | F_{t-1} &\sim N(0, \sigma_t^2), \\ \sigma_t^2 &= \gamma + \alpha Y_{t-1}^2, \end{aligned}$$

where  $F_{t-1}$  is the information available up to and including time  $t-1$ . We assume that  $0 \leq \alpha < 1$ , and therefore the process is weakly stationary. Assume also that  $Y_t^4$  is weakly stationary and  $3\alpha^2 < 1$  and notice that  $E(\varepsilon_t^4 | F_{t-1}) = 3$ , where  $\varepsilon_t = Y_t / \sigma_t$ . Show that

$$E(Y_t^4) = \frac{3\gamma^2}{1-3\alpha^2} \left( \frac{1+\alpha}{1-\alpha} \right).$$

2. Let  $x_t$  follow a GARCH(1,1) process, such that  $x_t = \sigma_t \varepsilon_t$ ,  $\sigma_t > 0$  with

$$\sigma_t^2 = \alpha_0 + \alpha_1 x_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

and  $\varepsilon_t \sim i.i.d.(0, 1)$  independent of  $\sigma_t$ . Let  $\sigma_h^2(\ell) = E(x_{h+\ell}^2 | F_h)$  where  $F_h$  is the information available up to and including time  $h$ . denote  $\sigma^2$  the unconditional variance of  $x_t$ . Show that if  $(\alpha_1 + \beta_1) < 1$

$$\sigma_h^2(\ell) = \sigma^2 + (\alpha_1 + \beta_1)^{\ell-1} (\sigma_h^2(1) - \sigma^2)$$

and consequently

$$\lim_{\ell \rightarrow \infty} \sigma_h^2(\ell) = \sigma^2.$$

[**Hint-** Recall the formula for sum of the numbers in a geometric progression:

$$\sum_{i=0}^n c^i = \frac{1 - c^{n+1}}{1 - c}$$

for a constant  $c$  and an integer  $n$ .]